

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	The formula is only valid when the angle AOB is given in radians	B1	This mark is given for a correct explanation
(b)	$\frac{40}{360} \times \pi \times 5^2$	M1	This mark is given for a correct method to find the area of the sector
	$\frac{25\pi}{9} \text{ cm}^2$	A1	This mark is given for a correct value for the area of the sector
(Total 3 marks)			

Q2.

Question	Scheme	Marks	AOs
(a)	Angle $AOB = \frac{\pi - \theta}{2}$	B1	2.2a
		(1)	
(b)	Area = $2 \times \frac{1}{2} r^2 \left(\frac{\pi - \theta}{2} \right) + \frac{1}{2} (2r)^2 \theta$	M1	2.1
	$= \frac{1}{2} r^2 \pi - \frac{1}{2} r^2 \theta + 2r^2 \theta = \frac{3}{2} r^2 \theta + \frac{1}{2} r^2 \pi = \frac{1}{2} r^2 (3\theta + \pi) *$	A1*	1.1b
		(2)	
(c)	Perimeter = $4r + 2r \left(\frac{\pi - \theta}{2} \right) + 2r\theta$	M1	3.1a
	$= 4r + r\pi + r\theta$ or e.g. $r(4 + \pi + \theta)$	A1	1.1b
		(2)	
(5 marks)			

Notes

(a)

B1: Deduces the correct expression for angle AOB

Note that $\frac{180-\theta}{2}$ scores B0

(b)

M1: Fully correct strategy for the area using their angle from (a) appropriately.

Need to see $2 \times \frac{1}{2} r^2 \alpha$ or just $r^2 \alpha$ where α is their angle in terms of θ from

part (a) + $\frac{1}{2} (2r)^2 \theta$ with or without the brackets.

A1*: Correct proof. For this mark you can condone the omission of the brackets in $\frac{1}{2} (2r)^2 \theta$ as

long as they are recovered in subsequent work e.g. when this term becomes $2r^2 \theta$

The first term must be seen expanded as e.g. $\frac{1}{2} r^2 \pi - \frac{1}{2} r^2 \theta$ or equivalent

(c)

M1: Fully correct strategy for the perimeter using their angle from (a) appropriately

Need to see $4r + 2r\alpha + 2r\theta$ where α is their angle from part (a) in terms of θ

A1: Correct simplified expression

Note that some candidates may change the angle to degrees at the start and all marks are available e.g.

$$(a) \frac{180 - \frac{180\theta}{\pi}}{2}$$

$$(b) 2 \left(\frac{180 - \frac{180\theta}{\pi}}{2} \right) \times \frac{1}{360} \times \pi r^2 + \frac{\theta}{360} \times \frac{180}{\pi} \times \pi (2r)^2 = \frac{1}{2} \pi r^2 - \frac{1}{2} r^2 \theta + 2r^2 \theta = \frac{1}{2} r^2 (3\theta + \pi)$$

$$(c) 4r + 2 \left(\frac{180 - \frac{180\theta}{\pi}}{2} \right) \times \frac{1}{360} \times 2\pi r + \frac{180\theta}{\pi} \times \frac{1}{360} \times 2\pi (2r) = 4r + \pi r + r\theta$$

Q3.

Question	Scheme	Marks	AOs
(a)	Uses $s = r\theta \Rightarrow 3 = r \times 0.4$	M1	1.2
	$\Rightarrow OD = 7.5 \text{ cm}$	A1	1.1b
		(2)	
(b)	Uses angle $AOB = (\pi - 0.4)$ or uses radius is $(12 - '7.5')$ cm	M1	3.1a
	Uses area of sector $= \frac{1}{2}r^2\theta = \frac{1}{2} \times (12 - 7.5)^2 \times (\pi - 0.4)$	M1	1.1b
	$= 27.8 \text{ cm}^2$	A1ft	1.1b
		(3)	
(5 marks)			
Notes:			
(a)			
M1: Attempts to use the correct formula $s = r\theta$ with $s = 3$ and $\theta = 0.4$			
A1: $OD = 7.5 \text{ cm}$ (An answer of 7.5cm implies the use of a correct formula and scores both marks)			
(b)			
M1: $AOB = \pi - 0.4$ may be implied by the use of $AOB = \text{awrt } 2.74$ or uses radius is $(12 - \text{their '7.5'})$			
M1: Follow through on their radius $(12 - \text{their } OD)$ and their angle			
A1ft: Allow awrt 27.8 cm^2 . (Answer 27.75862562). Follow through on their $(12 - \text{their '7.5'})$			
Note: Do not follow through on a radius that is negative.			

Q4.

Question	Scheme	Marks	AOs
	States or uses $\frac{1}{2}r^2\theta = 11$	B1	1.1b
	States or uses $2r + r\theta = 4r\theta$	B1	1.1b
	Attempts to solve, full method $r = \dots$	M1	3.1a
	$r = \sqrt{33}$	A1	1.1b
			[4]
(4 marks)			

Notes:

B1: States or uses $\frac{1}{2}r^2\theta = 11$ This may be implied with an embedded found value for θ

B1: States or uses $2r + r\theta = 4r\theta$ or equivalent

M1: Full method to find $r = \dots$ This involves combining the equations to eliminate θ or find θ
 The initial equations must be of the same "form" (see **) but condone slips when attempting to solve.

It cannot be scored from impossible values for θ Hence only score if $0 < \theta < 2\pi$ FYI $\theta = \frac{2}{3}$ radians

Allow this to be scored from equations such as $\dots r^2\theta = 11$ and ones that simplify to $\dots r = \dots r\theta$ **

Allow their $2r + r\theta = 4r\theta \Rightarrow \theta = \dots$ then substitute this into their $\frac{1}{2}r^2\theta = 11$

Allow their $2r + r\theta = 4r\theta \Rightarrow r\theta = \dots$ then substitute this into their $\frac{1}{2}r^2\theta = 11$

Allow their $\frac{1}{2}r^2\theta = 11 \Rightarrow \theta = \frac{22}{r^2}$ then substitute into their $2r + r\theta = 4r\theta \Rightarrow r = \dots$

A1: $r = \sqrt{33}$ only but isw after a correct answer.

The whole question can be attempted using θ in degrees.

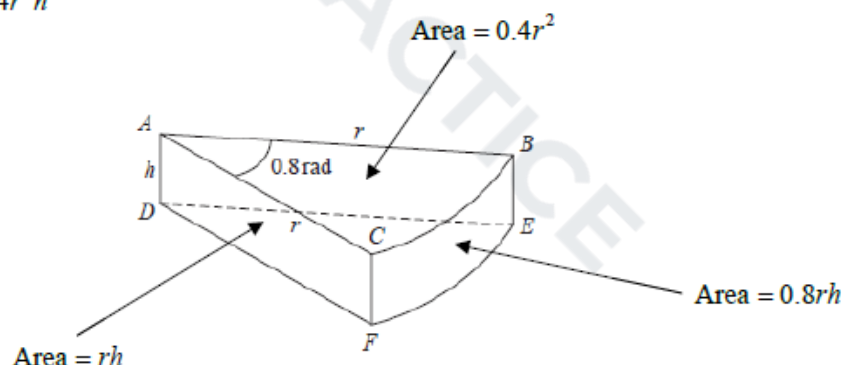
B1: States or uses $\frac{\theta}{360} \times \pi r^2 = 11$

B1: States or uses $2r + \frac{\theta}{360} \times 2\pi r = 4 \times \frac{\theta}{360} \times 2\pi r$

Q5.

Question	Scheme	Marks	AOs
(a)	Sets up an allowable equation using volume = 240 E.g. $\frac{1}{2}r^2 \times 0.8h = 240 \Rightarrow h = \frac{600}{r^2}$ o.e.	M1 A1	3.4 1.1b
	Attempts to substitute their $h = \frac{600}{r^2}$ into $(S =) \frac{1}{2}r^2 \times 0.8 + \frac{1}{2}r^2 \times 0.8 + 2rh + 0.8rh$	dM1	3.4
	$S = 0.8r^2 + 2.8rh = 0.8r^2 + 2.8 \times \frac{600}{r} = 0.8r^2 + \frac{1680}{r}$ *	A1*	2.1
		(4)	
(b)	$\left(\frac{dS}{dr}\right) = 1.6r - \frac{1680}{r^2}$	M1 A1	3.1a 1.1b
	Sets $\frac{dS}{dr} = 0 \Rightarrow r^3 = 1050$ $r = \text{awrt } 10.2$	dM1 A1	2.1 1.1b
		(4)	
(c)	Attempts to substitute their positive r into $\left(\frac{d^2S}{dr^2}\right) = 1.6 + \frac{3360}{r^3}$ and considers its value or sign	M1	1.1b
	E.g. Correct $\frac{d^2S}{dr^2} = 1.6 + \frac{3360}{r^3}$ with $\frac{d^2S}{dr^2} \Big _{r=10.2} = 5 > 0$ proving a minimum value of S	A1	1.1b
		(2)	
(10 marks)			
Notes:			

Volume = $0.4r^2h$



Total surface area = $2rh + 0.8r^2 + 0.8rh$

(a)

M1: Attempts to use the fact that the volume of the toy is 240 cm^3

Sight of $\frac{1}{2}r^2 \times 0.8 \times h = 240$ leading to $h = \dots$ or $rh = \dots$ scores this mark

But condone an equation of the correct form so allow for $kr^2h = 240 \Rightarrow h = \dots$ or $rh = \dots$

A1: A correct expression for $h = \frac{600}{r^2}$ or $rh = \frac{600}{r}$ which may be left unsimplified.

This may be implied when you see an expression for S or part of S E.g. $2rh = 2r \times \frac{600}{r^2}$

dM1: Attempts to substitute their $h = \frac{a}{r^2}$ o.e. such as $hr = \frac{a}{r}$ into a correct expression for S

Sight of $\frac{1}{2}r^2 \times 0.8 + \frac{1}{2}r^2 \times 0.8 + rh + rh + 0.8rh$ with an appropriate substitution

Simplified versions such as $0.8r^2 + 2rh + 0.8rh$ used with an appropriate substitution is fine.

A1*: Correct work leading to the given result.

$S =$, $SA =$ or surface area = must be seen at least once in the correct place

The method must be made clear so expect to see evidence. For example

$S = 0.8r^2 + 2rh + 0.8rh \Rightarrow S = 0.8r^2 + 2r \times \frac{600}{r^2} + 0.8r \times \frac{600}{r^2} \Rightarrow S = 0.8r^2 + \frac{1680}{r}$ would be fine.

(b) There is no requirement to see $\frac{dS}{dr}$ in part (b). It may even be called $\frac{dy}{dx}$.

M1: Achieves a derivative of the form $pr \pm \frac{q}{r^2}$ where p and q are non-zero constants

A1: Achieves $\left(\frac{dS}{dr}\right) = 1.6r - \frac{1680}{r^2}$

dM1: Sets or implies that their $\frac{dS}{dr} = 0$ and proceeds to $mr^3 = n$, $m \times n > 0$. It is dependent upon a

correct attempt at differentiation. This mark may be implied by a correct answer to their $pr - \frac{q}{r^2} = 0$

A1: $r = \text{awrt } 10.2$ or $\sqrt[3]{1050}$

(c)

M1: Attempts to substitute their positive r (found in (b)) into $\left(\frac{d^2S}{dr^2}\right) = e \pm \frac{f}{r^3}$ where e and f are non zero and finds its value or sign.

Alternatively considers the sign of $\left(\frac{d^2S}{dr^2}\right) = e \pm \frac{f}{r^3}$ (at their positive r found in (b))

Condone the $\frac{d^2S}{dr^2}$ to be $\frac{d^2y}{dx^2}$ or being absent, but only for this mark.

A1: States that $\frac{d^2S}{dr^2}$ or $S'' = 1.6 + \frac{3360}{r^3} = \text{awrt } 5 > 0$ proving a minimum value of S

This is dependent upon having achieved $r = \text{awrt } 10$ and a correct $\frac{d^2S}{dr^2} = 1.6 + \frac{3360}{r^3}$

It can be argued without finding the value of $\frac{d^2S}{dr^2}$. E.g. $\frac{d^2S}{dr^2} = 1.6 + \frac{3360}{r^3} > 0$ as $r > 0$, so

minimum value of S . For consistency it is also dependent upon having achieved $r = \text{awrt } 10$

Do NOT allow $\frac{d^2y}{dx^2}$ for this mark